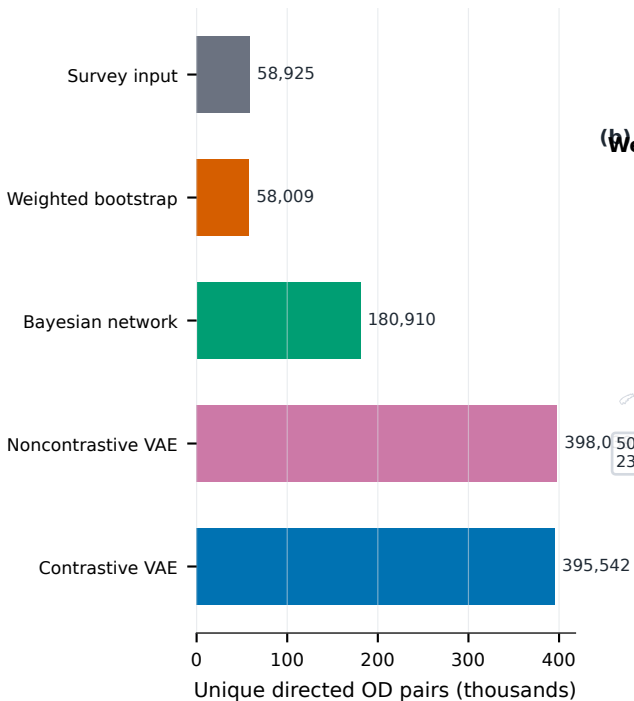


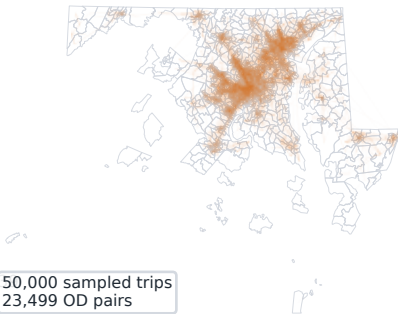
OD support expansion and tract-centroid desire lines

The VAE variants generate many more directed tract-pair combinations while the map uses equal-size row samples.

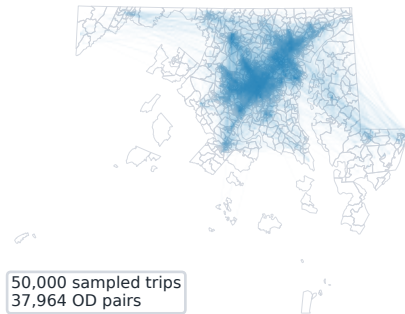
(a)



(b) Weighted bootstrap



Contrastive VAE



Lines are straight connections between tract representative points and do not indicate assigned routes.